

Mohammad Afzal Shadab

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Hydrology | Cryosphere | Modeling | Climate Change | Physics-Informed Machine Learning

POSITIONS HELD

Future Faculty in the Physical Sciences Postdoctoral Fellow Departments of Civil and Environmental Engineering and Geosciences, <i>Princeton University</i>	2024- Present Princeton
Graduate Research Assistant Oden Institute for Computational Engineering and Sciences, <i>University of Texas at Austin</i>	2019-24 Austin
NASA-JPL Graduate Fellow Planetary Science Division, <i>NASA Jet Propulsion Laboratory, California Institute of Technology</i>	Summer 2023 Pasadena
NASA-JPL Graduate Fellow Earth Science Division, <i>NASA Jet Propulsion Laboratory, California Institute of Technology</i>	Spring 2022 Pasadena
MIT Visiting Graduate Student Researcher Department of Mechanical Engineering, <i>Massachusetts Institute of Technology</i>	2018-19 Cambridge
Graduate Research Assistant Dept. of Mechanical and Aerospace Engg., <i>Hong Kong University of Science and Technology</i>	2016-18 Hong Kong

EDUCATION

Doctor of Philosophy Computational Science, Engineering & Mathematics The University of Texas at Austin, United States <i>Title:</i> Modeling Subsurface Flow of Water in Earth and Planetary Sciences <i>Advisor:</i> Dr. Marc Hesse, Professor of Earth and Planetary Sciences	2024
Master of Science Computational Science, Engineering & Mathematics The University of Texas at Austin, United States <i>Advisor:</i> Dr. Marc Hesse, Professor of Earth and Planetary Sciences	2021 GPA: 3.90/4.0
Master of Philosophy Mechanical Engineering The Hong Kong University of Science and Technology <i>Thesis:</i> Fifth-Order Finite Volume WENO in General Orthogonally-Curvilinear Coordinates  <i>Advisor:</i> Dr. Kun Xu, Chair Professor of Mathematics and Mechanical & Aerospace Engg.	2018 GPA: 4.0(A)/4.3(A+)
Bachelor of Technology Mechanical Engineering Aligarh Muslim University, India	2016 GPA: 9.62/10.0

GRANTS

Sustaining the Community Firm Model - NASA ROSES'24 Support for Open-Source Tools, Frameworks, and Libraries Collaborator (implementing enthalpy formation in CFM and validating), \$0 to PU, 1/2025-1/2028
Assessing Challenges for Polar Early Career Scientists During Science Policy Upheaval - PSECCO Polar Partnership Networking Event Collaboration Funding Support Co-PI, \$4000 to PU, 7/2025-6/2026
Carbon Dioxide Removal through Enhanced Rock Weathering Deployments with Smallholder Rice Paddy Farmers in India - Milkywire Climate Transformation Fund  Collaborator (on coupled hydrologic & reactive transport modeling) - \$0 to PU, 4/2025-7/2026
Oxidant Transport into Europa's Internal Ocean by Brine Migration Through the Outer Ice Shell - Research Award in Planetary Habitability by UT Center for Planetary Systems Habitability  PI - \$16,425, 08/2022-12/2022

RESEARCH EXPERIENCE

Modeling and Understanding Large-Scale Integrated Soil and Firm Hydrology	Princeton University
Postdoctoral Researcher; <i>Advisor</i> : Prof. Reed Maxwell, <i>Collaborator</i> : Prof. Howard Stone	Fall 2024- Present
Reactive Transport Modeling of Enhanced Weathering in Soils for CO₂ Removal	Princeton University
Collaborator; <i>Research Lead</i> : Dr. Jacob Jordan (Mati Carbon, XPRIZE'25 Carbon Removal ↗)	Summer 2024- Present
Modeling Subsurface Flow of Water in Earth and Planetary Sciences	UT Austin
Graduate Research Assistant, <i>Doctoral Thesis</i> ; <i>Advisor</i> : Prof. Marc Hesse	2019-24
Vadose Zone and Groundwater Hydrology on Early Mars	UT Austin
Research Lead; <i>Collaborators</i> : Dr. Eric Hiatt, Dr. R. Bahia (ESA), Dr. Eleni Bohacek (ESA)	2020-2024
Improving the Numerical Toolset for Geodynamics of Icy Oceans World	Jet Propulsion Lab
NASA Jet Propulsion Lab Graduate Fellow; <i>Advisor</i> : Dr. Steven Vance	Summer 2023
Modeling Meltwater Percolation in Greenland's Firm	Jet Propulsion Lab
NASA Jet Propulsion Lab Graduate Fellow; <i>Advisor</i> : Dr. Surendra Adhikari	Spring 2022
Investigating Groundwater Flows using Physics Informed Neural Networks 📄	UT Austin
Research Lead; <i>Collaborators</i> : Dr. Eric Hiatt, Dr. DingCheng Luo, Yiran Shen, Prof. Hesse	2020-23
Free Fall of a Viscous Drop in a Viscoelastic Medium ↗	Massachusetts Institute of Technology
Visiting Graduate Student Researcher; <i>Advisor</i> : Prof. Irmgard Bischofberger	2018-19
High-Order Finite-Volume Methods in Curvilinear Coordinates 📄	HKUST, Hong Kong
Graduate Research Assistant, <i>M.Phil. Thesis</i> ; <i>Advisor</i> : Prof. Kun Xu	2016-18

TEACHING EXPERIENCE (TA: TEACHING ASSISTANT, I: INSTRUCTOR, ALL FEEDBACK: [↗](#))

CEE 304 Energy and Environmental Engineering (Princeton U., Guest Evaluator)	Spring 2026
Wintersession 2024 Analyzing Remote Sensing Data with QGIS (Princeton U., I)	Winter 2024
GEO 325C/398C Continuum Mechanics ↗ (UT-Austin, TA)	Fall 2022, 2023
GEO 325M/398M Numerical Modeling in the Geosciences ↗ (UT-Austin, TA)	Spring 2023
SIAM Applied Mathematics Mentorship Program Lectures (UT-Austin, I)	Fall 2022
MECH 1907 Introduction to Aerospace Engineering (HKUST, TA)	Spring 2018
MECH 3690 Aerospace Engineering Laboratory (HKUST, TA)	Spring 2017

PEDAGOGICAL TRAINING

Teaching Transcript Program ↗ , Princeton University	Fall 2024-Spring 2026
Inclusive Course Design Institute 2023 ↗ , The University of Texas at Austin	Summer 2023
Inclusive Classrooms Leadership Certificate Seminar Series, University of Texas, Austin	Spring 2023
Advanced Teaching Preparation Series Certificate, University of Texas, Austin	Spring, Fall 2022
Graduate Teaching Assistant Training Program, H. K. University of Science and Technology	2017-18

ACCOLADES

Princeton University Future Faculty in the Physical Sciences Fellowship ↗	2024-27
Polar Science Early Career Community Office Polar Partnership Networking Award ↗	2025-26
AGU Cryosphere Innovation Award / Flash Freeze Competition Winner ↗	2023
UT Austin Graduate School Fellowships ↗	2021-22, Summer 2024, Spring 2024
Reviewing Recognitions by Geophysical Research Letters ↗ and J. of Geophysical Research ↗	2024
NASA Jet Propulsion Laboratory Graduate Fellowship ↗	Summer 2023, Summer 2022
European Space Agency Early Career Bursary Award ↗	2023
Purdue Climate Scholar by Purdue University and Office of Naval Research ↗	Summer 2022
MIT - Houston Energy Innovation Student Fellow ↗	2022-23
Lunar and Planetary Institute Career Development Award ↗	2022
Outstanding Student Presenters Award by Unsaturated Zone Technical Committee, AGU ↗	2021
Best Teaching Assistant Award by Dept. of Mech. & Aero. Engg., HKUST	2018
Outstanding Contribution in Reviewing Recognition by Journal of Computational Physics	2018
Postgraduate Research Scholarship (Studentship) by HKUST	2016-18

INVITED TALKS AND SEMINARS

- [12] January 2026: Stanford University, *Geophysics Department Seminar* [↗](#)
- [11] December 2025: AGU Fall Meeting, Session: *From Snowflakes to Runoff: Firn and Surface Mass Balance Processes*
- [10] September 2025: The University of New Mexico, *Department of Earth and Planetary Sciences Seminar*
- [9] April 2025: NASA Jet Propulsion Laboratory, *Earth Science Division Seminar*
- [8] April 2025: Princeton University, *Solid Earth Geosciences Brown Bag Seminar* [↗](#)
- [7] April 2024: Mathematics on Ice Forum [↗](#)
- [6] Jan 2024: California Institute of Technology - *Graduate Aerospace Laboratories (GALCIT) and Fu Research Group*
- [5] Oct 2023: The University of Texas at Austin - *Center for Planetary Systems Habitability*
- [4] Sept 2023: The University of Texas Institute for Geophysics
- [3] August 2023: NASA Jet Propulsion Laboratory, *Planetary Science Division Seminar*
- [2] June 2022: NASA Jet Propulsion Laboratory, *Earth Science Division Seminar*
- [1] June 2022: California Institute of Technology, *Fu Research Group*

SKILLS

Languages: C, C++, Fortran 77/90, Python (SciPy, NumPy, Matplotlib, Pandas, Pytorch, Tkinter, Tensorflow, GUI programming, Webscraping), HTML, Matlab, Mathematica, Shell Scripting, $\LaTeX$, High Performance Computing (SLURM), FEniCS

General Software: AutoCAD, SolidWorks, ANSYS, Fluent, COMSOL Multiphysics, TecPlot, ParaView, Microsoft Office, Git, Travis CI, Docker

Geoscience Software: Hydrus, Noah-MP, VPLanet, QGIS, QGreenland, ENVI, PlanetProfile, PHREEQC, ParFlow, Community Land Model, Community Firn Model

Remote Sensing & Gridded Products: GRACE/GRACE-FO, SMAP, GLDAS, ERA5 Reanalysis, OPERA-DISP, Sentinel-1, NISAR, OpenET, SSEBop, SAR/InSAR, raster/vector geospatial data

PEER REVIEWED PUBLICATIONS

- [13] **Shadab, M.A.**, Rutishauser, A., Grima, C. and Hesse, M.A., 2025. A unified kinematic wave theory for melt infiltration into firn. *Journal of Glaciology*, 71, e87, 25pp.
<https://doi.org/10.1017/jog.2025.10055>
- [12] **Shadab, M.A.**, Hiatt, E., Bahia, R.S., Bohacek, E.V., Steinmann, V. and Hesse, M.A., 2025. Infiltration dynamics on early Mars: Geomorphic, climactic, and water storage implications, *Geophysical Research Letters*, 52, e2024GL111939, 11+12pp.
<https://doi.org/10.1029/2024GL111939>
- [11] Barnes, R., ... **Shadab, M.A.**,..., 2025. History and habitability of the LP 890-9 planetary system, *The Planetary Science Journal*, 6(1), p.25, 13pp.
<https://www.doi.org/10.3847/PSJ/ad94dc>
- [10] Vanek, S., ..., **Shadab, M.A.**,..., 2024. Exploring the past, present, and future of USAPECS: Lessons from a decade of supporting early career research across national and international polar networks. *Arctic Yearbook*, 14pp.
<https://arcticyearbook.com/>
- [9] **Shadab, M.A.**, Adhikari, S., Rutishauser, A., Grima, C. and Hesse, M.A., 2024. A mechanism for ice layer formation in glacial firn. *Geophysical Research Letters*, 51(15), p.e2024GL109893, 12+37pp.
<https://doi.org/10.1029/2024GL109893>
- [8] **Shadab, M.A.** and Hesse, M.A., 2024. A hyperbolic-elliptic PDE model and conservative numerical method for gravity-dominated variably-saturated groundwater flow. *Advances in Water Resources*, p.104736, 17pp.
<https://doi.org/10.1016/j.advwatres.2024.104736>

- [7] Hiatt, E. **Shadab, M.A.**, Hesse, M., Goudge, T., Gulick, S., 2024. Limited recharge of the southern highlands aquifer on early Mars, *Icarus*, 408, p.115774, 10+16pp.
<https://doi.org/10.1016/j.icarus.2023.115774>
- [6] **Shadab, M.A.**, Hiatt, E. and Hesse, M.A., 2023. PKgui: A GUI software for Polubarinova-Kochina's solutions of steady unconfined groundwater flow, *SoftwareX*, 24, p.101573, 5+5pp.
<https://doi.org/10.1016/j.softx.2023.101573>
- [5] **Shadab, M.A.**, Luo, D., Hiatt, E., Hiatt, E. and Hesse, M.A., 2023. Investigating steady unconfined groundwater flow using physics informed neural networks, *Advances in Water Resources*, 177, p.104445, 16+18pp.
<https://doi.org/10.1016/j.advwatres.2023.104445>
- [4] **Shadab, M.A.** and Hesse, M.A., 2022. Analysis of gravity-driven infiltration with the development of a saturated region, *Water Resources Research*, 58(11), p.e2022WR032963, 27pp.
<https://doi.org/10.1029/2022WR032963>
- [3] **Shadab, M.A.**, Balsara, D., Shyy, W. and Xu, K., 2019. Fifth-order finite volume WENO in general orthogonally - curvilinear coordinates. *Computers & Fluids*, 190, 26pp.
<https://doi.org/10.1016/j.compfluid.2019.06.031>
- [2] **Shadab, M.A.**, Ji, X. and Xu, K., 2018. Fifth-order finite volume WENO on cylindrical grids. *Spectral and High Order Methods for Partial Differential Equations (Springer)*, 10pp.
https://doi.org/10.1007/978-3-030-39647-3_51
- [1] **Shadab, M.A.** and Baig, M.F., 2017. Investigation and control of unstart phenomenon in scramjets. In *21st AIAA International Space Planes and Hypersonics Technologies Conference* (p. 2298), 16pp.
<https://doi.org/10.2514/6.2017-2298>

UNDER REVIEW

- [7] **Shadab, M.A.**, Stone, H.A., and Maxwell, R.M., 202X. A vertically integrated model for aquifers in cold firn, arXiv:2510.14268. (Under revision in *Journal of Glaciology*)
- [6] **Shadab, M.A.**, Adhikari, S., Stevens, C.M., Rennermalm, A.K., Xiao, J., Hesse, M., and Maxwell, R.M., 202X. HydroFirn: A numerical model for large-scale multidimensional firn hydrology. arXiv:2604.11488 (Under revision in *Journal of Glaciology*)
- [5] **Shadab, M.A.**, Jadallah, N.S., and Maxwell, R.M., 202X. Effects of soil capillarity on multidimensional, integrated surface-subsurface hydrology at different spatial scales. (Under review in *Journal of Hydrology*)
- [4] Ishraque, F.*, **Shadab, M.A.***, Shionalyn, K., Narayanan, N., Schaberg, M., Gaikwad, S., Ezukanma, I., 202X. Assessing challenges for U.S.-based polar early career scientists during science policy upheaval. (Under review in *Journal of Glaciology*) *Co-first authors named alphabetically.
- [3] Hiatt, E. **Shadab, M.A.**, Hesse, M., Goudge, T., Gulick, S., 202X. Early Mars valley networks require decoupled surface and ground water. Research square: <https://doi.org/10.21203/rs.3.rs-10167991/v1> (Under review in *Nature Geoscience*)
- [2] Jordan, J.S., **Shadab, M.A.**, Prigiobbe, V., Kanzaki, Y., Planavsky, N., Reinhard, C., 202X. On the pH-dependent export of anthropogenic alkalinity in porewater through soil: Implications for enhanced rock weathering, CDRxiv: <https://doi.org/10.70212/cdrxiv.2026489.v1> (Under review in *American Journal of Science*)
- [1] Kiara P., Adams, K., Barge, L.M., Campubi, E., **Shadab, M.A.**, Teece, B.L., Daswani, M.M., Fraeman, A., and , Weber, J., 202X. Connections and Considerations for Application of Earth Science Models to Other Planetary Bodies: Mars as a Case Study. (Under review in *Planetary Science Journal*)

IN PREPARATION

- [7] **Shadab, M.A.**, Chand, M., Zeke, D., Morris, E., 202X. Geostatistical analysis of firn density variability along Greenland's EGIG Line at local and regional scales. (for *The Cryosphere*)
- [6] **Shadab, M.A.**, Vance, S.D., Silber, E A., Crósta, A.P., Carnahan, E., Jordan, J.S., Hesse, M.A., 202X. Rapid migration of impact melt through ocean world ices: Selk crater on Titan and Mannann'an crater on Europa. (for *Earth and Planetary Science Letters*)

- [5] **Shadab, M.A.** and Jordan, J.S., 202X. When does the alkalinity arrive? Wetting–drying controls on carbon dioxide removal by enhanced weathering. (for *Water Resources Research*)
- [4] **Shadab, M.A.**, Hiatt, E. and Hesse, M.A., 202X. Analytical solutions of a low-aspect-ratio unconfined aquifer on a spherical shell: Application to early Mars. (for *Journal of Geophysical Research - Planets*)
- [3] **Shadab, M.A.** and Hesse, M.A., 202X. An open source discrete operator toolbox (DOT) to solve geophysical flow problems. (for *Geoscientific Model Development*)
- [2] Hiatt, E. **Shadab, M.A.**, Gulick, S.P.S., Hesse, M.A., 202X. Experimental and numerical investigations of seepage face. (for *Journal of Fluid Mechanics*)
- [1] Hesse, M.A. and **Shadab, M.A.**, 202X. Numerical modeling for geoscientists. (book draft )

EXTENDED CONFERENCE ABSTRACTS

- [12] Hiatt, E., **Shadab, M.A.**, Gulick, S.P.S., Goudge, T. and Hesse, M.A., 2026. Transient Groundwater Modeling Suggests Extremely Punctuated Precipitation in an Early Mars Climate, *57th Lunar and Planetary Science Conference (LPSC)*, #1914, 2pp.
- [11] Hiatt, E., **Shadab, M.A.**, Gulick, S.P.S., Goudge, T. and Hesse, M.A., 2025. Transient Groundwater Recharge of Early Mars' Groundwater Systems & Subsequent Climate Constraints *56th LPSC*, #2629, 2pp.
- [10] **Shadab, M.A.**, Vance, S.D., Silber, E A., Crósta, A.P., Carnahan, E., Jordan, J.S., Hesse, M.A., 2024. Evolution of impact generated melt at Selk crater: Effect of phase change, percolation, and viscous foundering. *55th LPSC*, #1317, 2pp.
- [9] **Shadab, M.A.**, Hiatt, E., Bahia, R., Bohacek, E., Steinmann, V. and Hesse, M.A., 2024. Infiltration on early Mars & its implications toward aeolian-fluvial interactions. *55th LPSC*, #1383, 2pp.
- [8] Hiatt, E., **Shadab, M.A.**, Gulick, S.P.S., Goudge, T. and Hesse, M.A., 2024. Martian lakes: a critical requirement for transient groundwater models. *55th LPSC*, #2408, 2pp.
- [7] **Shadab, M.A.**, Hiatt, E. and Hesse, M.A., 2023. A deep crustal aquifer model for southern highlands of Noachian Mars shows groundwater age and near-surface dynamics. *NASA Exploration Science Forum*, 2pp.
- [6] **Shadab, M.A.**, Hiatt, E. and Hesse, M.A., 2023. A deep crustal aquifer model for southern highlands of Noachian Mars shows groundwater age and near-surface dynamics. *Brines Across the Solar System: Ancient and Future Brines Conference*, #2025, 2pp.
- [5] **Shadab, M.A.**, Hiatt, E. and Hesse, M.A., 2023. Investigating groundwater dynamics and residence times on early Mars using unconfined aquifer model with vertical heterogeneity. *54th LPSC*, #1736, 2pp.
- [4] Hesse, M.A., **Shadab, M.A.** and Hiatt, E., 2023. Time scales for terminal groundwater drainage from the southern highlands on Mars. *54th LPSC*, #1637, 2pp.
- [3] Hiatt, E., **Shadab, M.A.** and Hesse, M.A., 2023. Planetary scale groundwater and surface water interaction on early Mars. *54th LPSC*, #2415, 2pp.
- [2] **Shadab, M.A.**, Hiatt, E., and Hesse, M.A., 2022. Estimates of Martian mean recharge rates from analytic groundwater models. *53rd LPSC*, #1775, 2pp.
- [1] Hiatt, E., **Shadab, M.A.**, Gulick, S.P.S., Hesse, M.A., Goudge, T. and Hesse 2022. Estimates of groundwater divides and basins on Noachian Mars. *53rd LPSC*, #2618, 2pp.

SELECTED TALKS AND SEMINARS

- [9] **Shadab, M.A.**, 2026. Towards a mechanistic understanding of firn hydrology, *International Glaciological Society Global Seminar* .
- [8] **Shadab, M.A.** et al., 2026. Scale-Dependent Role of Soil Capillarity in Integrated Surface-Subsurface Hydrologic Simulations, *Groundwater Modeling and More*.
- [7] **Shadab, M.A.** et al., 2025. A theoretical & numerical model for unconfined aquifers in cold firn, *AGU Fall Meeting*.

- [6] **Shadab, M.A.** et al., 2024. Evolution of impact generated melt at Selk crater, *AGU Fall Meeting*.
- [5] **Shadab, M.A.** et al., 2024. Multi-scale multi dimensional infiltration in glacial firn and mechanism of ice layer and chunk formation, *AGU Fall Meeting*.
- [4] **Shadab, M.A.** et al., 2024. Dynamics of Infiltration on Early Mars, *AGU Fall Meeting*.
- [3] **Shadab, M.A.** et al., 2023. Mechanism & factors controlling ice layer formation in glacial firn, *AGU Fall Meeting*.
- [2] **Shadab, M.A.** et al., 2023. A unified kinematic wave theory for melt infiltration into firn, *AGU Fall Meeting*.
- [1] **Shadab, M.A.** et al., 2023. Infiltration on early Mars and its implications toward aeolian-fluvial interactions, *Fluvial-Aeolian Interactions on Planetary Surfaces (FAIRPLAY)*, *European Space Agency*.

OPEN SOURCE SOFTWARE

- [10] **Shadab, M.A.** et al., 2026. HydroFirn: A numerical model for large-scale multidimensional firn hydrology (v1.0.2). Zenodo. <https://doi.org/10.5281/zenodo.21672229>
- [9] **Shadab, M.A.** et al., 2025. ColdFirnAquifer3D: A Numerical Simulator for Aquifers in Cold Firn (v1.0). Zenodo. <https://doi.org/10.5281/zenodo.17354686>
- [8] **Shadab, M.A.** et al., 2025. Infiltration-on-early-Mars (v1.0.1). Zenodo. <https://doi.org/10.5281/zenodo.14742437>
- [7] **Shadab, M.A.** et al., 2024. unified-kinematic-wave-theory (v1.0). Zenodo. <https://doi.org/10.5281/zenodo.13936153>
- [6] **Shadab, M.A.** et al., 2024. mashadab/ice-layer-formation: v1.0.0, Zenodo. <https://doi.org/10.5281/zenodo.12706191>
- [5] **Shadab, M.A.** and Hesse, M. A., 2024. mashadab/VarSatFlow: v1.0, Zenodo. <https://doi.org/10.5281/zenodo.11398273>
- [4] **Shadab, M.A.** et al., 2023. mashadab/PKgui (v1.0.1), Zenodo. <https://doi.org/10.5281/zenodo.8034146>
- [3] **Shadab, M.A.** and Hesse, M.A., 2022. Gravity driven infiltration with the development of a saturated region (v1.0), Zenodo. <https://doi.org/10.5281/zenodo.6558260>
- [2] **Shadab, M.A.** et al., 2021. PINNs for unconfined groundwater flow (v1.0), Zenodo. <https://doi.org/10.5281/zenodo.5803542>
- [1] **Shadab, M.A.**, 2021. Reservoir-Simulator, Zenodo. <https://doi.org/10.5281/zenodo.6581752>

SERVICE

AGU26 Sessions' Convener, Four Sessions 	2026
Convening C036 - <i>Cryosphere Is for All</i> , P031 - <i>The New Mars Underground VIII</i> , C023 – <i>Monitoring and Modeling Wet Snow and Firn: Challenges, Opportunities, and Implications</i> , and Union Session U013 - <i>Navigating Broader Impacts in Current Political Climate</i> .	
AGU25 Sessions' Convener, Four Sessions 	2025
Convening C039 - <i>Cryosphere Is for All</i> , P041 - <i>The New Mars Underground VIII</i> , and Union Sessions C040 - <i>End of The Golden Era of Polar Science in the US?</i> , U014 - <i>Navigating Broader Impacts in Current Political Climate</i> .	
AGU24 Session Convener and OSPA Liaison and Judge 	2024
Designing oral/poster/e-lightening sessions with AGU Cryo team titled C24A/C41C/C43C <i>The Cryosphere Is for All: Overcoming Barriers to Participation in the Cryospheric Sciences at AGU24</i> .	
Executive Committee Member, AGU Cryosphere Division 	2024- Present
Serving in the Diversity, Equity, and Inclusion (DEI) and Canvassing Working groups.	Virtual

Judge, International Mission to Mars Engineering Design Contest ↗ Organized by Mars Society for high school students from around the world.	Summer 2024 Virtual
Co-Chair & DEI Team Lead, US Assoc. of Polar Early Career Scientists ↗ Fostering climate and DEI-conscious collaborations between academia & polar organizations.	2022- Present Virtual
Board Member, AGU Hydrology Section Student Subcommittee (AGU-H3S) ↗ Providing professional development & networking opportunities to early career hydrologists.	2023- Present Virtual
Volunteer, MIT Energy Conference ↗ Assisted in organizing in-person sessions at the conference.	2023 Boston
Coordinator, Center for Planetary Systems Habitability Student Travel Award ↗ Organized, coordinated and liaised the application process for student travel to LPSC 2023.	Spring 2023 Austin
MIT - Houston Energy Innovation Student Fellow ↗ Cultivated & supported energy innovation startup ecosystem considering threat of climate change.	2022-23 Austin
Volunteer / Braindate Lounge Assistant, AGU Fall Meeting 2022 ↗ Facilitated collaborations between researchers and scientists through Braindate at AGU 2022.	2022 San Francisco
Session Chair, Society for Industrial & Applied Mathematics Annual Meeting ↗ Chaired the <i>CP15: Machine Learning and Data Mining</i> Session.	2021 Virtual
President & Senior Advisor, SIAM Chapter of UT Austin ↗ Spearheaded several programs & won Best Graduate Organization at UT Austin Award.	2020-23 Austin
Volunteer, Lunar and Planetary Science Conference 2022 ↗ Managed a virtual session and an in-person session and helped with conference logistics.	2021 Houston

MENTORSHIP EXPERIENCE

Princeton High School Student Research Mentor (Phase I report) ↗ Mentoring Shaurya Ranjan on developing a physics informed machine learning model for wet firn hydrology, including data analysis, scientific writing, and presentations.	Fall 2025- Present Princeton
Interagency Arctic Research Policy Committee Mentorship program ↗ Providing career counseling and skills training.	2024- Present Virtual
Young Professional Mentor, Zed Factor Fellowship Program ↗ Mentored rising undergraduate students in aerospace engineering for skills development.	2023-24 Virtual
American Geophys. Union Earth & Planetary Surface Processes (EPSP) Mentorship ↗ Mentoring graduate students across the world to develop technical and research skills in EPSP.	2022- Present Virtual
Mentoring365, American Geophysical Union ↗ Facilitating an exchange of professional knowledge, skills, and experiences in Earth and space sciences.	2021- Present Virtual
SIAM Applied Mathematics Mentorship ↗ Founded the program and mentored undergrads for applied math concepts, research & careers.	2021-23 Austin
Sir Syed Global Scholar Award ↗ Mentoring top AMU students from humble backgrounds for US grad school applications.	2016- Present Virtual

OUTREACH

Science Outreach Educator, Integrated Ground Water Modeling Center, Princeton Univ. Instructed hydrologic modeling course to high schoolers; conducted outreach for K-12 students.	2024- Present
Team Member, UT Austin Libraries HELIOS team ↗ To advance Higher Education Leadership Initiative for Open Scholarship (HELIOS). Gave a speech at <i>US White House</i> Listening Sessions on Open Science (News ↗ , Post ↗ , Video ↗). Panelist at multiple open science events: Navigating the New Arctic ↗ , Texas Summit ↗ , IFLA ↗ .	2023-24 Austin

Geoscience Ambassador, Jackson School of Geosciences, UT Austin 	2021-22
Making geoscience accessible to broader community & promoting interdisciplinary research.	Austin
Zonal Head & College Head Ambassador, Smilyo Educational Charitable Society 	2014-15
Managed multi-university teams & provided educational resources to not-so-privileged.	New Delhi, India

REVIEWER

Journals (50+)

Geoscience: Nature Astronomy, Water Resources Res., Geophysical Research Lett., J. of Hydrology, Computational Geoscience, J. of Geophysical Research, Biosystems Engg., J. of Hydrometeorology, Discover Geoscience, J. of Applied Geophysics, IEEE Trans. on Geoscience and Remote Sensing, J. of Fluid Mechanics, Natural Hazards Res., Hydrology and Earth System Sciences (HESS)

Numerical Methods: J. of Computational Physics, Geoscientific Model Development, Computer and Fluids, Engineering with Computers, SoftwareX, Computers and Geosciences

Grant proposals (7)

NASA Research Opportunities in Space and Earth Sciences: Future Investigators in NASA Earth and Space Science and Technology (FINESST, Cryosphere), Early Career Investigator Program in Earth Science, Mars Data Analysis Program (2), Habitable Worlds, Hera Participating Scientist Program, Solar System Workings

Awards and Fellowships (7)

AGU Student Travel Award (2024, 2025), AGU Cryosphere Caregiver Award (2025), AGU Cryosphere CryoStuD Award (2024, 2025), Sir Syed Ahmed Global Scholar Award (2023, 2024)

Policy documents (2)

IPCC Seventh Assessment Report: Working Group I on Physical Science Basis for First Order Draft (2026), Second Order Draft (2026)

CERTIFICATIONS

EarthScope Consortium: InSAR Processing and Analysis (ISCE+) Technical Course (2026)

NASA: Open Science 101 (2024), NASA ARSET Monitoring Groundwater Changes for Water Management (2026)

CUAHSI: Snow Measurement Field Work (2026)

IEEE: Introduction to Geospatial Raster and Vector Data with Python (2024), Introduction to Radar Interferometry and Its Applications (2024), Deep Learning for Remote Sensing Data Analysis (2024)

CITI Program: IRB Social and Behavioral Research Investigators (2025)

MEDIA COVERAGE

\$50 million XPRIZE carbon removal awarded to Mati - Time  , Washington Post  , AP news 	2025
Is Mars Storing its Water Underground? - Universe Today  , Royal Astronomical Society 	2025
SciTech Daily  , Astrobiology  , UT  , AAAS  , Phys.org  , Earth.com  .	
Frankfurter Neue Presse  , TZ  , Ingenieur  , and Stars, Cells, and God Podcast  .	
Understanding ice layer formation to estimate sea level rise - UT  , AAAS  , Phys.org 	2024
History and Habitability of the LP 890-9 Planetary System - Astrobiology 	2024
Little groundwater recharge in ancient Mars aquifer - UT  , EurekAlert AAAS  , Phys.org 	2024
Fulfilling my NASA dream - Sir Syed Global Scholar Award Story of the Month 	2022
Mars may have less water than previously estimated - Multiple news outlets	2022
UT Austin Website (front cover)  ,  &  , Phys.org  , Times of India  , Bailey Universe 	
Outstanding Student Presentation Award at AGU 2021 - UT Austin 	2022
CPSH Travel Grant sends 11 students to LPSC - UT Austin 	2022
Lunar & Planetary Institute Career Devel. Award News - UT Austin  , LPI News 	2022
How to stay productive while in quarantine - Oden Institute Feature Article 	2021